

QUANTUM COMPUTING

PRINCIPLES, ALGORITHMS, INDUSTRIAL APPLICATIONS, AND ECONOMIC IMPACT

Exploring how quantum technologies are revolutionizing industries, solving complex problems, and shaping the global economy.



QUANTUM
PRINCIPLES



ADVANCED
ALGORITHMS



REAL-WORLD
APPLICATIONS



ECONOMIC
IMPACT



MD MONIRUZZAMAN
13-07-2025

“ Harnessing the laws of quantum mechanics to solve the world's most complex problems. ”

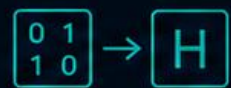
AGENDA / PRESENTATION OVERVIEW

A journey from the fundamentals of quantum computing to real-world impact with quantum annealing.



OUR GOAL

Understand how D-Wave quantum annealing solves complex optimization problems, with a focus on the Travelling Salesman Problem (TSP) and real industrial applications.

1 INTRODUCTION TO QUANTUM COMPUTING  Build the foundation: what is quantum computing and why it matters. • What is Quantum Computing? • Why Quantum Computing? • Bits vs Qubits • Superposition • Entanglement • Quantum Annealing Fundamentals	2 D-WAVE & QUANTUM ANNEALING  Explore D-Wave technology, its architecture, timeline, and position in the quantum landscape. • D-Wave Architecture • D-Wave Timeline • D-Wave vs IBM & Google • Comparison of Quantum Algorithms	3 WHY OPTIMIZATION MATTERS  Why optimization problems are crucial and where quantum can provide an advantage. • Why Optimization Matters • Quantum vs Classical Computing Performance	4 THE TRAVELLING SALESMAN PROBLEM  Understand TSP, its complexity, classical approaches, and real-world applications. • What is TSP? • Why TSP is NP-Hard? • Why is TSP Difficult? • Real-World Applications • Classical Approaches • How Quantum Helps	5 FROM TSP TO QUANTUM SOLUTION  Transform TSP into a form quantum annealers can solve and walk through the complete workflow. • TSP to QUBO • QUBO to Hamiltonian • Quantum Annealing Workflow • Live Example: 6-City TSP	6 INDUSTRIAL APPLICATIONS  See how quantum optimization is already creating impact in real industries. • Case Study: Deutsche Bahn • Other Companies Using Quantum Optimization • Real Industrial Workflow	7 CHALLENGES & FUTURE OUTLOOK  Current limitations, research challenges, and the future of quantum optimization. • Current Limitations • Research Challenges • Future Outlook	8 CONCLUSION  Key takeaways and final thoughts. • Key Takeaways • References • Questions & Discussion
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WHAT TO EXPECT



Clear concepts and simple explanations



Real technology: D-Wave quantum annealer



Step-by-step TSP to quantum solution workflow



Real-world case studies and impact



Honest look at challenges and the road ahead

WHAT IS QUANTUM COMPUTING?



BUILT ON QUANTUM MECHANICS

Unlike classical computers that use bits (0 or 1), quantum computers use quantum bits or qubits, which can exist in multiple states simultaneously.



SUPERPOSITION

Qubits can be in a combination of 0 and 1 at the same time, allowing quantum computers to process a vast number of possibilities in parallel.



ENTANGLEMENT

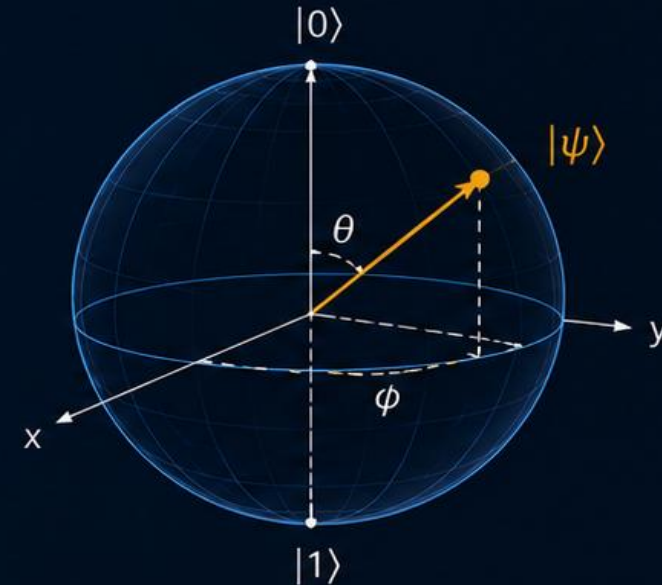
Qubits can be entangled, meaning the state of one qubit is correlated with the state of another, no matter the distance.



POWERING THE NEXT COMPUTING REVOLUTION

Quantum computing is ideal for solving problems that are intractable for classical computers, such as simulation, optimization, and cryptography.

THE QUBIT ON THE BLOCH SPHERE



$|0\rangle$ = Classical 0
(North Pole)

$|\psi\rangle$ = A combination
of 0 and 1

$|1\rangle$ = Classical 1
(South Pole)



Quantum computers leverage superposition and entanglement to explore many solutions simultaneously—unlocking new capabilities across industries.



SIMULATION



OPTIMIZATION



CRYPTOGRAPHY

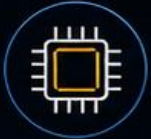


MACHINE LEARNING

Quantum computing is not just faster computing—it's fundamentally more powerful.

WHY NOW?

THE PERFECT CONVERGENCE OF TECHNOLOGY,
INVESTMENT, AND NEED



HARDWARE BREAKTHROUGHS

Rapid advancements in qubit technology, coherence times, and qubit counts are bringing fault-tolerant, practical quantum computers closer than ever.

Google, IBM, IonQ, Rigetti, D-Wave



CLOUD ACCESSIBILITY

Quantum computers are now accessible via the cloud, lowering the barrier for researchers, developers, and businesses.

AWS Braket, Microsoft Azure Quantum, IBM Quantum



QUANTUM ADVANTAGE EMERGING

Early demonstrations, including Google's quantum supremacy (2019), mark the beginning of quantum advantage for real-world problems.

Beyond classical limits for specific tasks



STRONG INVESTMENT & GOVERNMENT SUPPORT

Billions of dollars are being invested globally by governments and private sectors to lead the next computing revolution.

US, EU, China, Canada, Japan, India, Australia



GROWING ECOSYSTEM & TALENT

A thriving ecosystem of startups, academic research, and industry partnerships, along with a surge in quantum-skilled professionals.

Engineers, Scientists, Developers, Innovators

FROM THEORY TO TRANSFORMATION

Quantum computing is moving from labs to industries, unlocking solutions to problems once thought impossible.



2010–2015

Foundational
research and early
qubit prototypes



2016–2018

Rise of quantum
cloud platforms
and increased
investments



2019

Google claims
quantum
supremacy



2020–2023

Rapid hardware
improvements and
real-world pilot
projects



2024+

Toward quantum
advantage in
industry-scale
applications



We are at the starting line of the quantum era—
the time to understand, build, and lead is now.



QUANTUM COMPUTING FUNDAMENTALS

THE BUILDING BLOCKS OF THE QUANTUM WORLD

Quantum computers harness the strange and powerful laws of quantum mechanics to process information in fundamentally new ways.

1. BITS vs QUBITS

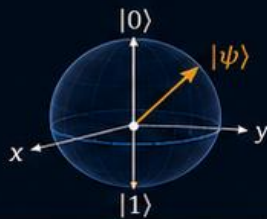
CLASSICAL BIT

Can be only 0 or 1



QUBIT

Can be 0, 1, or both at the same time



A qubit exists in a *superposition* of states, enabling massive parallelism.

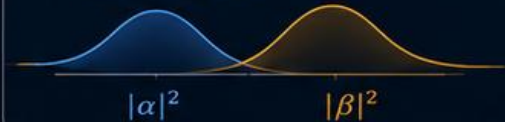
2. SUPERPOSITION

A qubit can be in a combination of $|0\rangle$ and $|1\rangle$.

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

$$\text{where } |\alpha|^2 + |\beta|^2 = 1$$

The measurement yields 0 with probability $|\alpha|^2$ and 1 with probability $|\beta|^2$.



3. ENTANGLEMENT

Qubits can become entangled, meaning the state of one instantly influences the other, no matter the distance.



Bell State Example:

$$\frac{|00\rangle + |11\rangle}{\sqrt{2}}$$

Measuring one qubit instantly determines the state of the other.

4. QUANTUM GATES

Quantum gates manipulate qubits, similar to logic gates in classical computing.

HADAMARD (H)

Creates superposition



$$\frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

PAULI-X (X)

Bit flip



$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

CNOT

Entangling gate



5. QUANTUM CIRCUITS

Quantum algorithms are built using sequences of quantum gates.



6. MEASUREMENT

Measurement collapses the qubit state to either $|0\rangle$ or $|1\rangle$.



7. DECOHERENCE

Interaction with the environment causes loss of quantum information.



8. THE NISQ ERA

Today's quantum computers are Noisy Intermediate-Scale Quantum (NISQ) devices.



50 – 1000+ Qubits



Accessible via the Cloud



Useful for specific tasks



Bridge to fault-tolerant future

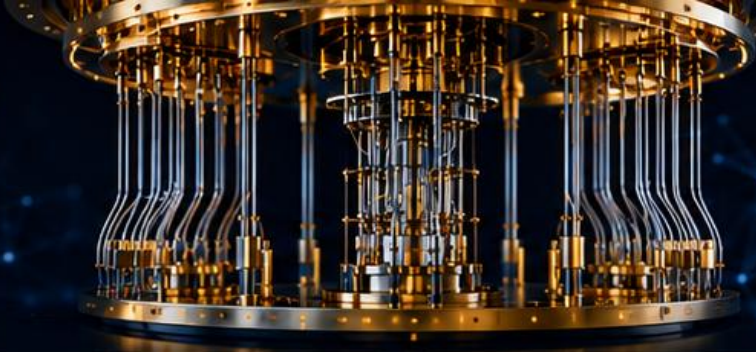


These principles—**superposition**, **entanglement**, and **interference**—enable quantum computers to solve certain problems much faster than classical computers.

QUANTUM HARDWARE

THE ENGINES OF QUANTUM COMPUTING

Different physical systems can be used to build qubits. Each technology has unique advantages, challenges, and use cases.



1. SUPERCONDUCTING QUBITS



- Most widely used technology today.
- Qubits are made from superconducting circuits cooled near absolute zero.
- Fast gate operations.
- Challenged by decoherence and require complex cryogenic systems.

LEADING COMPANIES



2. ION TRAP QUBITS



- Ions are trapped and manipulated using electromagnetic fields.
- Very high qubit fidelity.
- Excellent coherence times.
- Slower gate speeds compared to superconducting qubits.

LEADING COMPANIES



3. PHOTONIC QUBITS

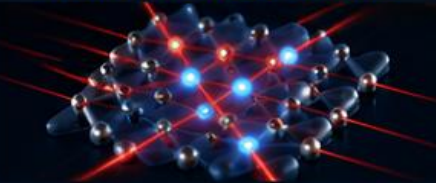


- Uses photons (light particles) to encode qubits.
- Operates at room temperature.
- Low decoherence and easy networking.
- Challenged by photon loss and deterministic operations.

LEADING COMPANIES



4. NEUTRAL ATOM QUBITS



- Atoms are trapped in optical lattices using laser beams.
- Highly scalable and reconfigurable.
- Promising for large-scale systems.
- Still in early development stage.

LEADING COMPANIES



5. TOPOLOGICAL QUBITS (FUTURE)



- Encodes information in global properties of the system.
- Naturally resistant to certain types of errors.
- Theoretical advantage for fault tolerance.
- Still purely experimental.

LEADING RESEARCH INITIATIVES



COMPARISON OVERVIEW	Technology	Qubit Quality	Gate Speed	Scalability	Operating Temp.	Maturity	Best Suited For
	Superconducting	★★★★☆	★★★★★	★★★☆☆	~10 mK	High	Near-term applications
	Ion Trap	★★★★★	★★★☆☆	★★★☆☆	~10 mK	Medium	High-fidelity computing
	Photonic	★★★★☆	★★★★★	★★★★☆	Room Temp.	Medium	Quantum communication
	Neutral Atoms	★★★★☆	★★★☆☆	★★★☆☆	Room Temp.	Low	Large-scale systems
	Topological (Future)	★★★★★	★★★☆☆	★★★★★	Unknown	Very Low	Fault-tolerant future

💡 No single technology is “best” for everything—different hardware platforms will drive quantum computing forward in complementary ways.

QUBITS: THE QUANTUM BIT

THE BASIC UNIT OF QUANTUM INFORMATION

While classical computers use bits that are either 0 or 1, quantum computers use **qubits**, which can exist in multiple states simultaneously.

CLASSICAL BIT

Can be either 0 or 1

0

State 0

1

State 1

VS

QUBIT

Can be in a *superposition* of 0 and 1



$|\psi\rangle$

Can be in only one state at a time.

Can be in multiple states simultaneously.

KEY PROPERTIES OF QUBITS



SUPERPOSITION

Allows qubits to be in multiple states at once, enabling massive parallelism.



ENTANGLEMENT

Qubits become correlated so that the state of one instantly affects the other, no matter the distance.

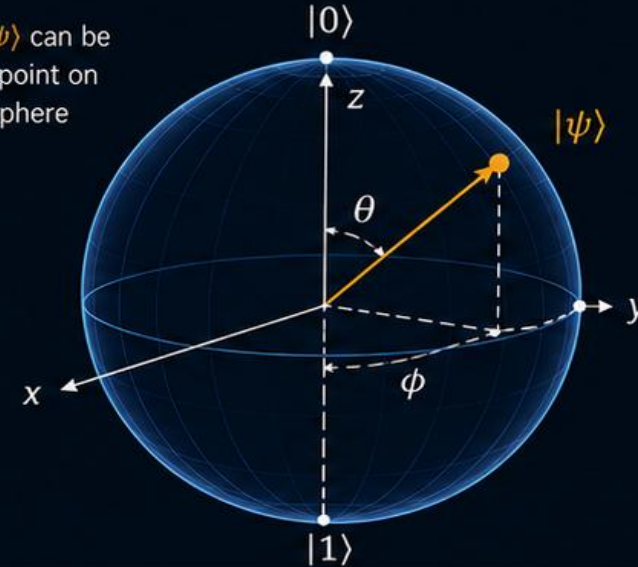


INTERFERENCE

Probability amplitudes can constructively or destructively interfere, enhancing the correct answers.

THE BLOCH SPHERE REPRESENTATION

Any qubit state $|\psi\rangle$ can be represented as a point on the surface of a sphere (Bloch Sphere).



$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

$$|\alpha|^2 + |\beta|^2 = 1$$

(Normalization)

- θ (theta): angle from the z-axis (0 to π)
- ϕ (phi): angle in the x-y plane (0 to 2π)
- $|\alpha|^2$ = probability of measuring 0
- $|\beta|^2$ = probability of measuring 1

EXAMPLE:

$$\text{If } |\psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle), \text{ then } |\alpha|^2 = |\beta|^2 = \frac{1}{2}$$

The qubit has a 50% chance of being measured as 0 or 1.



MATHEMATICAL STATE

A general qubit state:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

where $\alpha, \beta \in \mathbb{C}$ and $|\alpha|^2 + |\beta|^2 = 1$

Measurement collapses $|\psi\rangle$ to:

- $|0\rangle$ with probability $|\alpha|^2$
- $|1\rangle$ with probability $|\beta|^2$

WHY QUBITS MATTER



Simulations of quantum systems



Solving complex optimization problems



Breaking current cryptography (and creating new secure protocols)



Enabling quantum machine learning and AI

SUPERPOSITION

THE POWER OF BEING IN MANY STATES AT ONCE

Superposition is the ability of a qubit to be in 0 and 1 at the same time until it is measured.

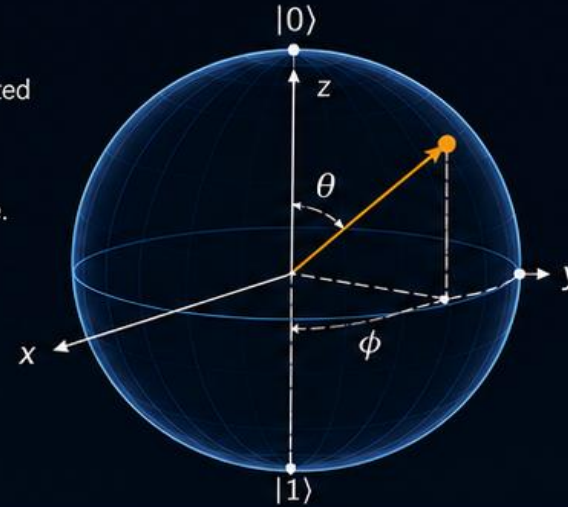


Classical bits are like switches. Qubits are like dimmer switches that can be anywhere between 0 and 1.



VISUALIZING SUPERPOSITION ON THE BLOCH SPHERE

A qubit's state can be represented as a point on the surface of the Bloch sphere.



General qubit state:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

where:

$\alpha, \beta \in \mathbb{C}$ (complex numbers)

Normalization condition:

$$|\alpha|^2 + |\beta|^2 = 1$$

- $|\alpha|^2$ = probability of measuring 0
- $|\beta|^2$ = probability of measuring 1

CLASSICAL BIT

Must be in one of the two states.



State 0



State 1

VS

QUBIT (SUPERPOSITION)

Can be in a combination of both states.



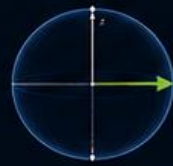
At any moment, it is either 0 OR 1.

It is 0 AND 1 at the same time until measured.

EXAMPLES OF SUPERPOSITION STATES

EQUAL SUPERPOSITION

$$\alpha = \frac{1}{\sqrt{2}}, \beta = \frac{1}{\sqrt{2}}$$

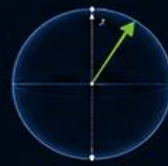


$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

50% chance of 0
50% chance of 1

MOSTLY 0

$$\alpha = 0.8, \beta = 0.6$$

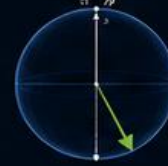


$$|\psi\rangle = 0.8|0\rangle + 0.6|1\rangle$$

64% chance of 0
36% chance of 1

MOSTLY 1

$$\alpha = 0.3, \beta = 0.95$$



$$|\psi\rangle = 0.3|0\rangle + 0.95|1\rangle$$

9% chance of 0
91% chance of 1

SUPERPOSITION LEADS TO INTERFERENCE

Multiple paths in superposition can interfere—amplifying correct answers and canceling wrong ones.

CONSTRUCTIVE INTERFERENCE

Waves add up.



Amplitude increases

DESTRUCTIVE INTERFERENCE

Waves cancel out.



Amplitude decreases (cancels)



KEY TAKEAWAY

Superposition allows qubits to process a vast number of possibilities simultaneously.
This is the foundation of quantum speedup.



ENTANGLEMENT

A MYSTERIOUS CONNECTION BEYOND DISTANCE

Entanglement is a unique quantum phenomenon where two or more qubits become linked, so that the state of one instantly influences the other, no matter how far apart they are.



“ Entanglement is not just spooky — it’s a resource.”

BELL STATE EXAMPLE

A pair of qubits can be entangled in a Bell state:

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}} (|00\rangle + |11\rangle)$$



When one qubit is measured, the other instantly collapses to the correlated state.

HOW ENTANGLEMENT WORKS

- 1 Two qubits are prepared in an entangled state.
- 2 They are sent far apart.
- 3 Measuring one qubit randomly gives 0 or 1.
- 4 The other qubit instantly becomes the opposite state.

The results are perfectly correlated, no matter the distance.

KEY FEATURES

- No classical explanation**
Entanglement cannot be explained by classical physics.
- Instant correlations**
Correlations appear instantly, but no information is transmitted faster than light.
- Stronger than classical**
Entangled states enable tasks impossible for classical systems.
- Fragile**
Easily destroyed by noise and environmental interactions.

VISUALIZING ENTANGLEMENT



Measure left qubit: 0 → Right qubit: 0



Measure left qubit: 1 → Right qubit: 1



Outcomes are random, but perfectly correlated.

REAL-WORLD IMPACT

- Quantum Cryptography**
Provides unbreakable communication (QKD).
- Quantum Networking**
Enables future quantum internet and repeaters.
- Quantum Computing**
Improves algorithms and error correction.
- Quantum Sensing**
Enhances precision in measurement technologies.

EXAMPLE TECHNOLOGIES

- IBM Quantum Systems (Eagle, Osprey)
- IonQ Trapped Ion Systems (Aria, Forte)
- Xanadu Photonic Chips (Borealis)
- PsiQuantum Photonic Qubits (Fault-tolerant vision)



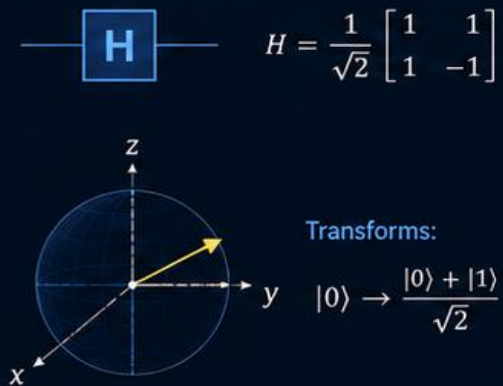
QUANTUM GATES

THE BUILDING BLOCKS OF QUANTUM CIRCUITS

Quantum gates are operations that change the state of qubits. They are the quantum equivalent of logic gates in classical computers.

HADAMARD GATE (H)

Creates superposition



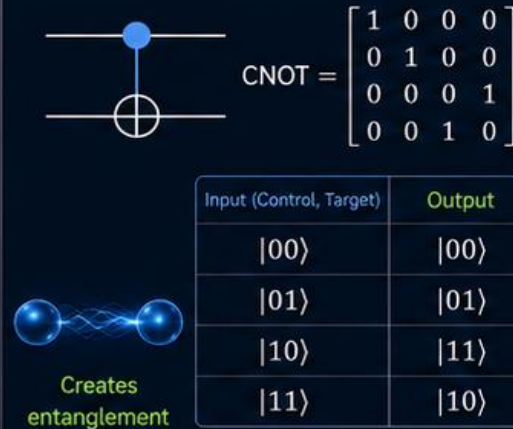
PAULI-X GATE (X)

Bit flip (NOT gate)



CNOT GATE

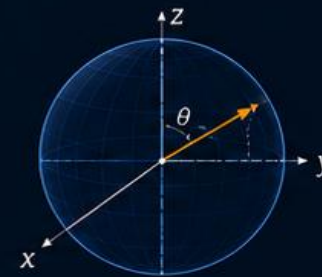
Entangling two qubits



UNIVERSAL SET OF GATES

Any quantum circuit can be built using a universal set of gates, such as $\{H, X, CNOT\}$.

Example: Rotate qubit around X -axis by angle θ



$$R_x(\theta) = e^{-i\theta X/2}$$
$$= \cos \frac{\theta}{2} I - i \sin \frac{\theta}{2} X$$

Matrix form:

$$R_x(\theta) = \begin{bmatrix} \cos(\theta/2) & -i \sin(\theta/2) \\ -i \sin(\theta/2) & \cos(\theta/2) \end{bmatrix}$$



COMPUTATION

Enable powerful quantum algorithms beyond classical limits.



SECURITY

Power cryptographic protocols like QKD for unbreakable security.



COMMUNICATION

Support quantum repeaters and future quantum internet.



SIMULATION

Model complex quantum systems in chemistry and materials.



OPTIMIZATION

Solve real-world optimization problems more efficiently.



Quantum gates are the tools that turn **qubits** into a powerful computing resource—enabling the **quantum advantage**.

APPLICATIONS & THE FUTURE

WHERE QUANTUM COMPUTING CAN MAKE A DIFFERENCE

Quantum computing has the potential to solve problems that are impossible or impractical for classical computers.



Quantum computing won't replace classical computers—it will complement them and **unlock new possibilities.**

DRUG DISCOVERY & MATERIALS SCIENCE



Simulate molecules and reactions at the quantum level.

Example:
Faster discovery of new medicines and advanced materials.

OPTIMIZATION PROBLEMS



Solve complex optimization problems much faster.

Example:
Logistics, supply chain, scheduling, and traffic management.

FINANCE & RISK ANALYSIS



Analyze large datasets and model risk more accurately.

Example:
Portfolio optimization, fraud detection, and option pricing.

CRYPTOGRAPHY & SECURITY



Break current encryption methods and create quantum-safe security.

Example:
Post-quantum cryptography for future-proof data protection.

ARTIFICIAL INTELLIGENCE & MACHINE LEARNING



Speed up AI training and solve complex learning tasks.

Example:
Quantum-enhanced machine learning algorithms.

THE FUTURE IS QUANTUM



Rapid Progress
Quantum hardware is improving every year.



Hybrid Future
Quantum and classical computers will work together.



New Industries
Entire industries will be transformed by quantum technologies.



Global Race
Countries and companies worldwide are investing heavily in quantum research.

CHALLENGES AHEAD



QUBIT STABILITY

Qubits are extremely sensitive to noise and environmental interference.



ERROR RATES

Quantum systems have higher error rates and need advanced error correction.



HIGH COST

Building and maintaining quantum computers is expensive.



LIMITED QUANTUM ALGORITHMS

We are still in the early stages of developing practical quantum algorithms.



KEY TAKEAWAY

Quantum computing is not just a technological advancement—it is a new paradigm.

It has the potential to solve humanity's most complex problems and shape the future.

While challenges remain, the potential benefits of quantum computing are **enormous**—and the journey has only **just begun**.



D-WAVE QUANTUM ANNEALER ARCHITECTURE

Specialized Hardware for Optimization

D-Wave quantum annealers are not universal quantum computers. They are purpose-built to solve optimization problems by finding low-energy states of an Ising Hamiltonian using quantum annealing.



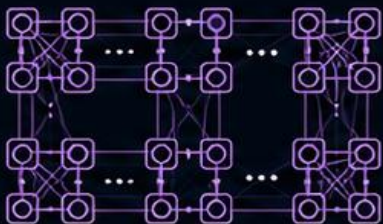
AT A GLANCE

- Thousands of qubits
- Sparse qubit connectivity
- Works at millikelvin temperatures
- Finds (near-)optimal solutions to QUBO problems



TOPOLOGY EVOLUTION

CHIMERA (DW2000Q)



- Unit cell: 8 qubits
- Sparse connections within and between cells
- Limited connectivity
- Up to ~2,000 qubits

PEGASUS (DW2000Q™ and Advantage)



- Higher connectivity
- Reduced number of required qubits for embedding
- Up to ~5,000 qubits (Advantage)

ZEPHYR (Advantage2)



- Even higher connectivity
- Better performance and embedding efficiency
- 20+ qubit connectivity
- Up to 7,000+ qubits (Advantage2)

HOW IT WORKS: QUANTUM ANNEALING

1. Initialize



All qubits start in a superposition of states.

2. Anneal



A quantum process slowly evolves the system toward lower energy states.

3. Quantum Fluctuations



Quantum tunneling helps the system escape local minima to find better solutions.

4. Measure



The system is measured, returning a (near-)optimal solution.

KEY ARCHITECTURAL FEATURES



Superconducting Qubits
Qubits are implemented using superconducting loops with Josephson junctions.



Dilution Refrigerator
Operates at ~15 millikelvin to maintain superconductivity and reduce thermal noise.



Sparse Connectivity
Not all qubits are connected. Embedding maps logical problems onto physical hardware.



Designed for Optimization
Specifically built to solve QUBO and Ising problems efficiently.

FROM PROBLEM TO HARDWARE



Problem Formulation
Define the optimization problem as a QUBO.



Embedding
Map the logical problem onto hardware topology.



Quantum Annealing
Run the annealing process on the D-Wave system.



Solution
Retrieve the best (lowest energy) solution found.



Why this matters: The architecture is engineered to efficiently explore a huge number of possible solutions in parallel, making it ideally suited for hard combinatorial optimization problems like the Travelling Salesman Problem.

D-WAVE
The Quantum Computing Company™

D-WAVE TIMELINE

Over Two Decades of Innovation in Quantum Optimization

From an idea to a leader in quantum annealing systems, D-Wave continues to push the boundaries of what's possible.

“

Our mission is to build quantum computers that solve real-world problems.

– D-Wave Systems



1999

D-Wave Founded

D-Wave Systems is founded in Vancouver, Canada with a vision to build quantum computers that can solve real-world problems.



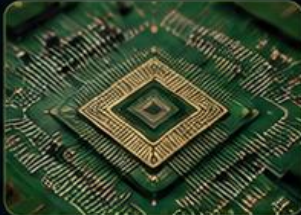
The journey begins



2007

First Commercial Prototype

D-Wave One™ prototype is built — the first quantum computer based on quantum annealing.



Proof of concept



2011

First Commercial Quantum Computer

D-Wave One™ is released commercially — the world's first quantum computer available to customers.



Quantum computing becomes real



2015

D-Wave 2X™

1000+ qubits and major performance improvements in coherence time and energy scale.



Stronger and more reliable systems



2020

Advantage™ System

Introduced Advantage — with 5000+ qubits and hybrid solvers to tackle bigger optimization problems.



Built for real-world impact



2023

Advantage2™ Prototype

Next-generation architecture with higher qubit count, better connectivity, and improved performance.



Pushing performance to the next level



The Future

Scaling to millions of qubits, improving accuracy, and solving increasingly complex real-world challenges across industries.



Toward practical quantum advantage



Key Milestones

From a bold idea to real-world quantum systems used by enterprises and researchers around the globe.



20+ Years
of quantum
innovation



100+
Employees
worldwide



500+
Customers &
partners



30+
Industries
impacted



Millions
of problems
solved



Our Commitment

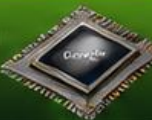
D-Wave continues to innovate and collaborate with industries, researchers, and governments to unlock the full potential of quantum optimization.

HOW D-WAVE COMPARES TO IBM AND GOOGLE

Different Approaches, Same Goal: Solving Real-World Problems with Quantum Power

“ There is no one universal quantum computer—different architectures excel at different tasks.
– Quantum Computing Reality

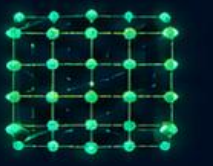


Aspect	D-WAVE (Quantum Annealing) 	IBM QUANTUM (Gate-Based) 	GOOGLE QUANTUM (Gate-Based) 
 Core Approach	Quantum Annealing Specialized for optimization problems	Gate-Based Universal Quantum Computing Can run many types of algorithms	Gate-Based Universal Quantum Computing Focus on error correction and scalability
 Best For	- Combinatorial Optimization - Scheduling - Routing, Logistics, TSP	- Chemistry Simulation - Machine Learning - Cryptography Research	- Large-Scale Computation - Quantum Error Correction - Scientific Breakthroughs
 Qubit Type	Superconducting Qubits (Annealing Qubits)	Superconducting Qubits (Gate Model)	Superconducting Qubits (Gate Model)
 Current Strengths	Thousands of qubits (Advantage2: 7000+ qubits) High qubit count for optimization	High gate fidelity Wide ecosystem (Qiskit, Cloud) Strong research & development	Breakthrough in Quantum Error Correction (Surface Code) Leading in scientific impact
 Accessibility	Commercial Cloud Access (Leap™ platform)	IBM Quantum Cloud (Open to developers & researchers)	Google Quantum Cloud (Research partners, limited access)
 Commercial Focus	Enterprise & Industry (Optimization at scale)	Hybrid Cloud & Research (General-purpose quantum)	Research-Driven (Scaling & error correction)
 Example Systems	Advantage / Advantage2 (7,000+ qubits)	IBM Quantum Eagle (127+ qubits) IBM Quantum Heron (133+ qubits)	Sycamore, Willow (53–100+ qubits)

At a Glance: Key Differences

-  **D-Wave = Specialist**
Built for optimization. Best choice when your problem is an optimization challenge.
-  **IBM = Versatile**
General-purpose quantum computers. Great for many different quantum algorithms.
-  **Google = Pioneer**
Pushing the boundaries of quantum science, error correction, and scalability.

Quick Visual Comparison

D-WAVE	IBM	GOOGLE
Optimization (Find the best route)	Universal Computation (Solve many problems)	Quantum Breakthroughs (Scale & accuracy)
		

When to Choose What?

-  **Choose D-Wave if...**
You have a real-world optimization problem like routing, scheduling, resource allocation, or TSP. 
-  **Choose IBM if...**
You want flexibility, experimentation, and a wide range of quantum algorithms and simulations. 
-  **Choose Google if...**
You are researching the future of quantum computing and working on error correction or scale. 

 **Bottom Line**
D-Wave, IBM, and Google are not competitors in the same way—they are pioneers in **different paths** toward the **quantum future**. Together, they drive the entire quantum ecosystem forward.

COMPARISON OF QUANTUM ALGORITHMS

Different Algorithms for Different Superpowers

Quantum computing offers a variety of algorithms, each designed to solve specific types of problems more efficiently than classical approaches.

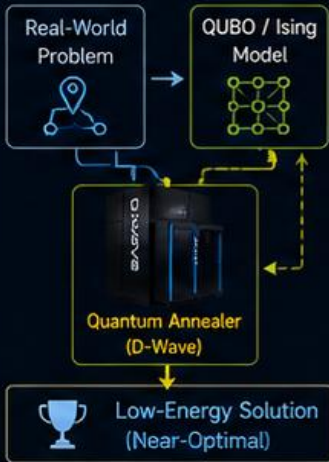


No single algorithm solves everything.
Each quantum algorithm has unique strengths and is best suited for specific problem domains.

Algorithm	 Shor's Algorithm	 Grover's Algorithm	 QAOA (Quantum Approximate Optimization Algorithm)	 VQE (Variational Quantum Eigensolver)	 Quantum Annealing (D-Wave Approach)
 Best For	Factoring large numbers	Searching unsorted databases	Combinatorial optimization	Quantum chemistry & materials	Combinatorial optimization (large-scale)
 Problem Type	Number theory	Search problems	Optimization problems (e.g., routing, scheduling)	Finding ground state of molecules	Optimization problems (e.g., TSP, vehicle routing)
 Quantum Advantage	Exponential speedup	Quadratic speedup	Potential advantage (varies by problem)	Potential advantage for chemistry problems	Heuristic speedup on certain optimization problems
 Output	Exact factors	Marked item(s)	Near-optimal solution	Estimated ground state energy	Near-optimal solution
 Maturity	Theoretical (not practical yet at scale)	Theoretical (small-scale demos exist)	Early-stage (active research)	Early-stage (active research)	Commercially available today (D-Wave systems)
 Example Use Cases	Cryptography breaking (RSA)	Database search, pattern matching	MaxCut, portfolio optimization, routing	Drug discovery, molecular simulation	TSP, scheduling, supply chain, logistics

WHERE D-WAVE FITS

D-Wave systems use quantum annealing to find low-energy states of Ising models, which can be mapped to many real-world optimization problems.



KEY TAKEAWAY

Quantum algorithms are specialized tools. Choosing the right algorithm (or approach) depends on the problem you want to solve.



Shor
Breaks cryptography



Grover
Searches faster



QAOA
Optimizes combinatorial problems



VQE
Solves quantum chemistry



Quantum Annealing (D-Wave)
Solves large-scale optimization problems

WHY OPTIMIZATION MATTERS

Optimization is About Making the Best Possible Decisions

At its core, optimization means finding the best solution from countless possibilities while satisfying certain constraints.



Goal: Achieve the best outcome (e.g., minimum time, cost, risk, distance) using available resources under real-world constraints.

EVERY DAY, OPTIMIZATION POWERS THE WORLD AROUND US



Saves Time



Reduces Cost



Improves Efficiency



Manages Risk



Lowens Energy & Emissions

REAL-WORLD EXAMPLES



1. GOOGLE MAPS ROUTES

Finds the fastest or shortest route considering traffic, road conditions, and real-time updates.

➡ Optimizes travel time and fuel consumption.



2. AMAZON DELIVERY

Optimizes delivery routes for thousands of packages while satisfying time windows and vehicle capacity.

➡ Reduces delivery time and operational cost.



3. AIRLINE SCHEDULING

Assigns aircraft and crews, schedules flights, and manages connections across hubs.

➡ Improves on-time performance and reduces delays.



4. PORTFOLIO OPTIMIZATION

Allocates investments across assets to maximize returns while controlling risk and meeting constraints.

➡ Balances risk and return for better financial outcomes.



5. MANUFACTURING

Optimizes production schedules, machine assignments, and supply chains under capacity constraints.

➡ Increases throughput and reduces waste.



THE BIG IDEA: Many real-world problems can be formulated as optimization problems. One of the most famous examples is the Travelling Salesman Problem (TSP).



QUANTUM vs CLASSICAL COMPUTING PERFORMANCE

Two Different Paradigms for Solving Problems

Classical computers process information sequentially, exploring one possibility at a time.
Quantum annealers explore many possibilities simultaneously, increasing the chance of finding better solutions.

FEATURE	CLASSICAL COMPUTING	QUANTUM ANNEALING
 Search Method	Sequential search Explores one solution path at a time	Parallel exploration Explores many solution paths simultaneously
 Processing	Deterministic Follows a defined set of rules	Probabilistic Leverages quantum fluctuations to escape local minima
 Speed	Depends on algorithm Time increases exponentially for many optimization problems	Potential exponential speedup Advantage for some optimization problems (e.g., TSP, scheduling)
 Solution	Exact or heuristic Can guarantee optimal solution (given enough time) or use approximations	Near-optimal Returns high-quality solutions quickly; not always the exact optimal solution
 Scalability	Limited by exponential growth Resource requirements grow rapidly with problem size	Better exploration of large solution spaces Designed to handle large-scale combinatorial optimization problems
 Best Use	General-purpose computing Wide range of applications (databases, AI, simulations, etc.)	Optimization problems Scheduling, routing, portfolio optimization, resource allocation, TSP, etc.

HOW THEY EXPLORE SOLUTIONS

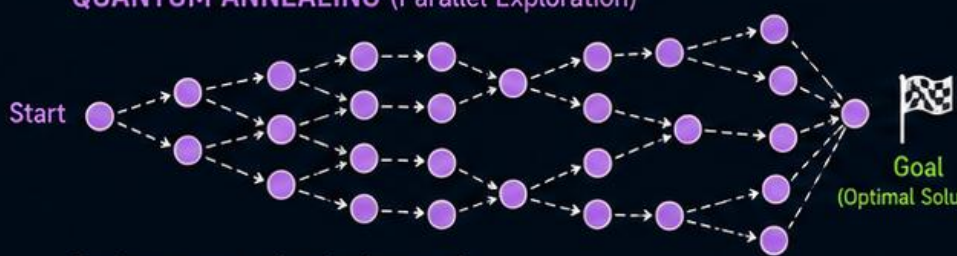
CLASSICAL COMPUTING (Sequential Search)



Start  →  →  →  →  →  →  → ... →  Goal (Optimal Solution)

Explores one path at a time.
May take a long time to find the best solution.

QUANTUM ANNEALING (Parallel Exploration)



Start  →  →  →  →  →  →  →  →  →  →  →  Goal (Optimal Solution)

Explores many paths simultaneously.
Higher chance of finding better (lower-energy) solutions faster.



KEY TAKEAWAY

Quantum annealing does not replace classical computers. It **complements** them by solving **specific optimization problems** that are too complex for classical computers to handle efficiently.



Quantum computing is not about being faster at everything.
It's about solving the **right problems** in a fundamentally better way.



Explore more possibilities



Find better solutions in less time



Handle larger problem sizes



Solve real-world optimization challenges

TRAVELLING SALESMAN PROBLEM (TSP)

WHAT IS IT?



The Travelling Salesman Problem (TSP) asks: Given a set of cities and the distances between them, what is the **shortest possible route** that **visits each city exactly once** and returns to the starting city?



THE GOAL

Find the **minimum total distance** (or cost) route that includes every city exactly once and returns to the starting city.



WHY IT MATTERS

TSP models many real-world routing and scheduling problems, such as delivery routes, logistics, and transportation planning.



TSP looks simple to understand, but solving it exactly becomes **extremely hard** as the number of cities increases.

EXAMPLE: 5 CITIES

Consider 5 cities with the distances shown below. The task is to find the shortest possible tour.



ONE POSSIBLE TOUR (NOT NECESSARILY THE SHORTEST)

A → B → C → D → E → A

Total distance

1,700 km

★ Our goal is to find the tour with the **smallest possible** total distance.

WHY TSP IS NP-HARD

The Computational Difficulty Behind the Problem

The Travelling Salesman Problem (TSP) becomes intractable as the number of cities increases. It is one of the most well-known NP-Hard problems.



IN SIMPLE TERMS

As the number of cities grows, the number of possible routes grows exponentially. Even the world's fastest supercomputers cannot check all possibilities in a reasonable time.



COMPLEXITY CLASSES EXPLAINED

P

P (Polynomial Time)

Problems that can be solved quickly (in polynomial time) by a computer.
Example: Sorting a list of numbers.

NP

NP (Nondeterministic Polynomial Time)

Problems for which a proposed solution can be verified quickly, even if finding the solution is hard.
Example: Given a route, we can check if it visits each city once and returns to the start.

NP-Hard

NP-Hard (At Least as Hard as NP)

Problems that are at least as hard as the hardest problems in NP. They may not even be in NP.
Example: TSP with distances that don't satisfy the triangle inequality.

WHY TSP BECOMES EXPONENTIALLY HARD



Number of Cities (n)

Number of Possible Tours ($(n-1)! / 2$)

5

12

10

181,440

15

87,178,291,200

20

6.08×10^{16}

25

1.55×10^{21}

30

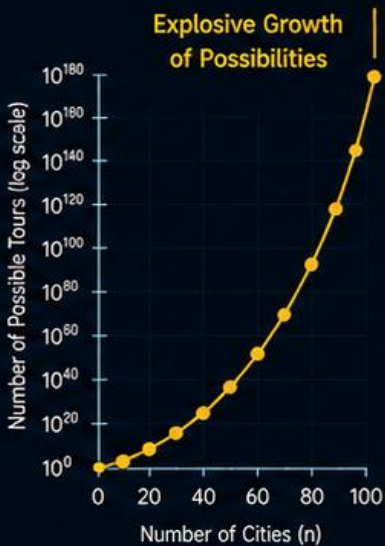
2.65×10^{25}

50

3.04×10^{62}

100

9.33×10^{156}



TSP IS NP-HARD



TSP can be shown to be **NP-Hard** by reducing other known NP-Hard problems to it.



This means if we had a fast algorithm for TSP, we could solve many other **NP-Hard problems** fast too.



No known algorithm can solve TSP exactly in polynomial time for all cases.



Therefore, we rely on **approximation algorithms**, **heuristics**, or emerging approaches like **quantum computing** to find near-optimal solutions.



IMPLICATIONS

- Exact solutions become impractical beyond small problem sizes.
- We need smarter algorithms, problem reformulations, and new technologies.
- Quantum computing offers a promising path to tackle such hard problems more efficiently.



KEY TAKEAWAY

TSP is NP-Hard, meaning the time required to find the exact best solution grows **exponentially** with the number of cities. This is why new computing paradigms like **quantum annealing** are so exciting.



WHY IS TSP DIFFICULT?

THE EXPLOSION OF POSSIBILITIES

TSP requires checking a massive number of possible routes to find the shortest one. The number of possibilities grows factorially with the number of cities.



NUMBER OF POSSIBLE TOURS (Fixing the starting city):

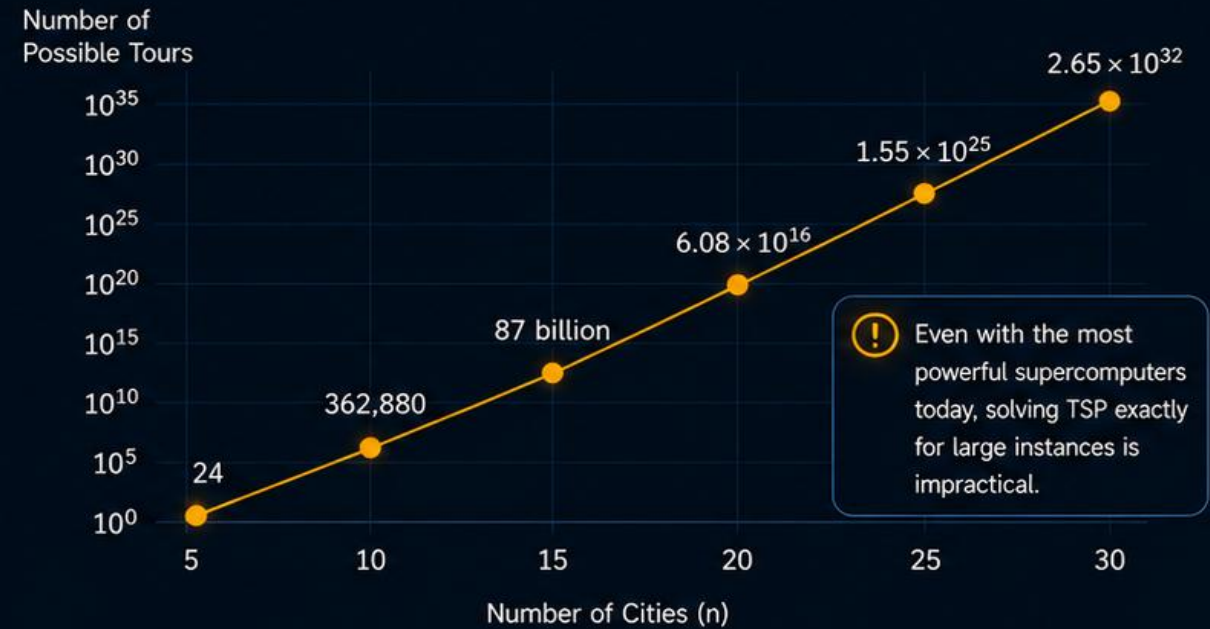
$$(n - 1)! \quad \text{where } n = \text{number of cities}$$

GROWTH OF POSSIBLE TOURS

Number of Cities (n)	5	10	15	20	25	30
Possible Tours ((n-1)!)	24	362,880	87,178,291,200	6.08×10^{16}	1.55×10^{25}	2.65×10^{32}
Approx. Time (1 billion tours/second)	0.000024 s	0.00036 s	87 s	1,927 years	4.9×10^9 years	8.4×10^{16} years

① Assuming a very fast computer checking 1 billion (10^9) tours per second.

POSSIBLE TOURS GROW FACTORIALLY



COMBINATORIAL EXPLOSION



KEY TAKEAWAY

The number of possible routes increases factorially. A small increase in the number of cities leads to an enormous explosion of possibilities.

This is why TSP is one of the most challenging problems in computer science.

REAL-WORLD APPLICATIONS OF TSP

From theory to impact – solving real optimization challenges

TSP appears in many industries where finding the most efficient route or schedule saves **time**, reduces **cost**, and improves **sustainability**.



TSP is not just a math problem — it is a real-world optimization challenge that **drives efficiency** in our daily lives.

• WHERE TSP MAKES A DIFFERENCE •



LOGISTICS & DELIVERY



Find the shortest delivery route that visits all locations and returns to the depot.

Example: DHL, UPS, Amazon



AIRLINE ROUTE PLANNING



Optimize flight routes to reduce fuel consumption, time, and operational cost.

Example: Lufthansa, Air India, Emirates



RAILWAY & PUBLIC TRANSPORT



Optimize train routes and schedules to reduce delays and improve network efficiency.

Example: Deutsche Bahn, Eurostar, Indian Railways



FIELD SERVICE & TECHNICIAN ROUTING



Plan the most efficient route for technicians to visit all service locations in a day.

Example: Siemens, Bosch, Vodafone



WAREHOUSE & ROBOT NAVIGATION

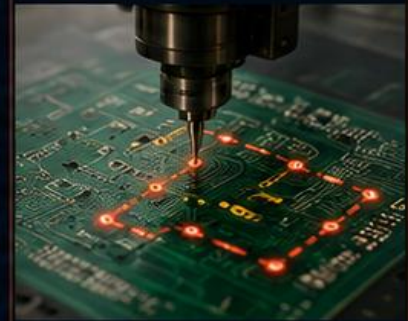


Determine the optimal picking path for robots to collect items and return to base.

Example: Amazon Robotics, Ocado, Alibaba



MANUFACTURING & PCB DRILLING



Find an optimal drilling or cutting path that visits all points with minimal movement.

Example: Siemens, Bosch, PCBWay



From global transportation networks to everyday deliveries, TSP helps industries move **smarter**, **faster**, and more **sustainably**.



Save Time



Reduce Cost



Lower Emissions



Improve Efficiency

CLASSICAL APPROACHES TO TSP

Powerful Algorithms, But They Hit a Wall

Many classical methods can find good solutions to TSP, but each has trade-offs in **speed**, **accuracy**, and **scalability**.



THE GOAL

Find the shortest possible tour visiting every city once and returning to the starting city.



1. BRUTE FORCE

Try Every Possible Tour



Enumerates all possible tours and picks the best one.



Advantages

Guaranteed optimal solution



Limitations

Factorial time complexity – impossible for large n

2. DYNAMIC PROGRAMMING

Break It Into Subproblems



Solves smaller subproblems and builds up to the final solution.



Advantages

Exact solution, better than brute force



Limitations

High time and memory usage – expensive for large n

3. GENETIC ALGORITHMS

Evolve Better Solutions



Uses evolution (selection, crossover, mutation) to improve solutions.



Advantages

Good for large problems, flexible and widely used



Limitations

No guarantee of optimality, tuning parameters is complex

4. ANT COLONY OPTIMIZATION

Learn From Pheromone Trails



Artificial ants build solutions and reinforce good paths with pheromones.



Advantages

Good for large, complex instances

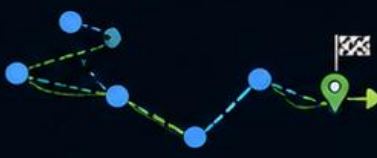


Limitations

Slower convergence, solution quality varies

5. GREEDY / HEURISTIC METHODS

Make Locally Best Choices



Builds a tour by repeatedly choosing the nearest (or best) next city.



Advantages

Very fast and simple



Limitations

Often far from optimal solution

COMPARISON OVERVIEW

Approach	Optimal Solution Guaranteed?	Time Complexity (Approx.?)	Scalability (Large n)	Typical Use Cases
Brute Force	Yes	$O(n!)$	Very Poor	Very small problems ($n \leq 10$)
Dynamic Programming	Yes	$O(n^2 \cdot 2^n)$	Poor	Medium-sized problems ($n \leq 20$)
Genetic Algorithms	No	$O(k \cdot n^2)$	Good	Large problems ($n \geq 50$)
Ant Colony Optimization	No	$O(k \cdot n^2)$	Good	Large, complex problems
Greedy / Heuristic	No	$O(n^2)$	Excellent	Very large problems, quick solutions

n = number of cities, k = number of iterations / population size



KEY TAKEAWAY

Classical algorithms have been extremely successful in solving TSP for decades.

However, as the number of cities grows, they either become too slow, too memory-intensive, or cannot guarantee the optimal solution.



This is where quantum approaches offer a new direction for optimization.

HOW QUANTUM COMPUTING HELPS

A New Paradigm for Solving Optimization Problems

Quantum computers use the principles of **superposition**, **entanglement**, and **quantum interference** to **explore** and evaluate **many possible solutions** at the same time.

This gives them a unique advantage for solving complex optimization problems like TSP.



Quantum computing does not replace classical computing — it **complements** it.

It offers new tools to solve problems that are **intractable** for classical computers.



• WHY OPTIMIZATION PROBLEMS BENEFIT •



SUPERPOSITION

Qubits can be in 0, 1, or both states simultaneously. Quantum computers can represent **many solutions** at once.



ENTANGLEMENT

Qubits become correlated, so the state of one affects others. This allows the system to capture **complex relationships** between decisions.



QUANTUM INTERFERENCE

Combined with entanglement, interference amplifies **good solutions** and cancels out bad ones, guiding the system toward optimal or near-optimal answers.



PARALLEL EXPLORATION

Quantum algorithms can evaluate a huge number of possibilities in parallel, offering potential **exponential speedup** for certain problem classes.

• QUANTUM VS CLASSICAL MINDSET •

CLASSICAL APPROACH

(One Path at a Time)



- Explores one solution at a time
- Checks possibilities **sequentially**
- Easy to implement
- Struggles as **problem size grows**

VS

QUANTUM APPROACH

(Many Paths at Once)



- Explores many solutions in **parallel**
- Uses **interference** to focus on better solutions
- Potential for **exponential speedup**
- Promising for large, complex optimization problems

• KEY QUANTUM ALGORITHMS FOR OPTIMIZATION •



QUANTUM ANNEALING

Uses quantum fluctuations to find the global minimum of a cost function. Well-suited for problems like TSP.



QUANTUM APPROXIMATE OPTIMIZATION ALGORITHM (QAOA)

A hybrid quantum-classical algorithm that uses variational optimization to approximate the best solution.



THE TAKEAWAY

- ✓ Quantum computing provides **new ways** to tackle optimization problems that are too hard for classical computers.
- ✓ For problems like TSP, quantum algorithms can explore a **massive** solution space more efficiently.
- ✓ While we are still in early stages, quantum optimization has the potential to **revolutionize industries** like logistics, transportation, and scheduling.



Quantum computing doesn't just compute faster — it **thinks differently**.



That difference could be the **key** to solving the **world's hardest optimization problems**.

CONVERTING TSP INTO A QUBO PROBLEM

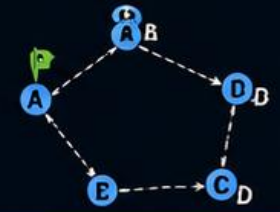
From Routes to a Quadratic Unconstrained Binary Optimization Model

QUBO (Quadratic Unconstrained Binary Optimization) is a way to encode optimization problems using **binary variables** and a **quadratic objective function** that a quantum annealer can solve.

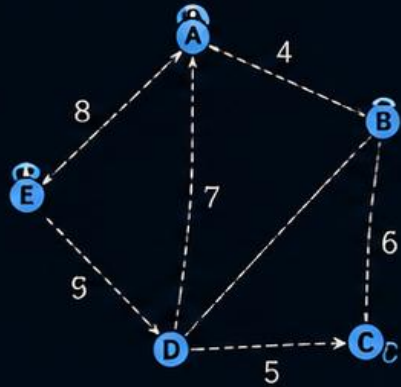


THE GOAL

Find the shortest Hamiltonian cycle (tour) that visits each city exactly once and returns to the starting city.



1 TSP EXAMPLE (5 CITIES)



Distances are symmetric.

Goal: Find the shortest tour starting and ending at A.

2 DEFINE DECISION VARIABLES

Let $x_{i,p} = \begin{cases} 1, & \text{if city } i \text{ is at position } p \\ 0, & \text{otherwise} \end{cases}$

For 5 cities and 5 positions, we have $5 \times 5 = 25$ binary variables.

		Positions in the tour (p)				
		1	2	3	4	5
Cities (i)	A	$x_{A,1}$	$x_{A,2}$	$x_{A,3}$	$x_{A,4}$	$x_{A,5}$
	B	$x_{B,1}$	$x_{B,2}$	$x_{B,3}$	$x_{B,4}$	$x_{B,5}$
	C	$x_{C,1}$	$x_{C,2}$	$x_{C,3}$	$x_{C,4}$	$x_{C,5}$
	D	$x_{D,1}$	$x_{D,2}$	$x_{D,3}$	$x_{D,4}$	$x_{D,5}$
	E	$x_{E,1}$	$x_{E,2}$	$x_{E,3}$	$x_{E,4}$	$x_{E,5}$

✓ Each city is visited exactly once:



$$\sum_{p=1}^5 x_{i,p} = 1 \quad \forall i$$

✓ Each position is filled by exactly one city:



$$\sum_{i \in \{A, B, C, D, E\}} x_{i,p} = 1 \quad \forall p$$

These are hard constraints in the QUBO model.

3 DEFINE THE COST FUNCTION

The cost is the total distance of the tour.

If city i is at position p and city j is at position $p+1$, we add distance $d_{i,j}$.

Total distance term:

$$C_{dist} = \sum_{p=1}^n \sum_i \sum_{j \neq i} d_{i,j} x_{i,p} x_{j,(p \bmod n)+1}$$

Add penalty terms for the constraints:

- Penalize if a city appears more than once
- Penalize if a position has more than one city

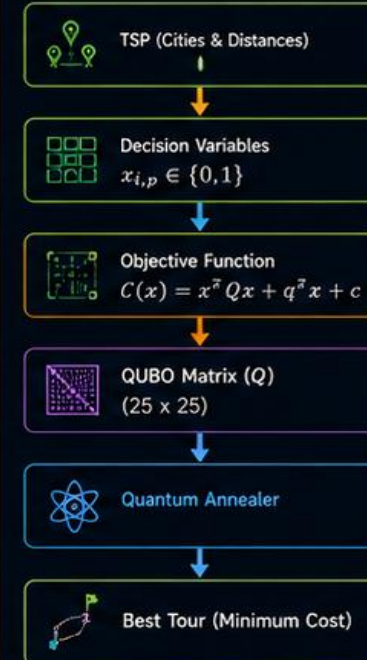
QUBO objective:

$$\text{Minimize } C(x) = C_{dist} + \lambda_1 C_{city} + \lambda_2 C_{pos}$$

where x is the vector of 25 binary variables and λ_1, λ_2 are large penalty coefficients.

i The objective $C(x)$ is quadratic (only x_i and $x_i x_j$ terms) and unconstrained $\rightarrow$ perfect for QUBO!

4 RESULT: A QUBO MODEL



★ WHY QUBO?

⚡ Captures all constraints and objective in a single mathematical model.

⚙️ Can be solved by quantum annealers that find low-energy (low-cost) solutions.

📈 Scales better for hard combinatorial problems compared to brute force.



IN SUMMARY

We transform the TSP into a QUBO by representing tour decisions with **binary variables**, building an objective function for total distance, and adding penalty terms for constraints. The resulting **quadratic model** can be minimized on a **quantum annealer** to find a (near-)optimal tour.

FROM QUBO TO HAMILTONIAN

Turning an Optimization Problem into an Energy Landscape

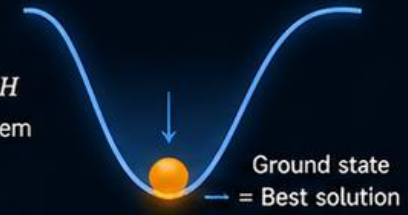
A quantum annealer does not solve the QUBO directly.

It solves an equivalent quantum system whose energy is described by a **Hamiltonian**.



THE GOAL

Map the QUBO objective function to a Hamiltonian H so that the lowest energy state of the quantum system corresponds to the optimal tour.



1 QUBO OBJECTIVE

We start with the QUBO objective:

$$\text{Minimize } C(x) = x^T Q x + q^T x + c$$

where:

- x is a binary vector (0 or 1)
- Q is a symmetric matrix (quadratic terms)
- q is a linear coefficient vector
- c is a constant



This function encodes the total distance and penalty terms of the TSP.

2 EXPAND THE QUBO

The QUBO can be written as:

$$C(x) = \sum_i q_i x_i + \sum_{i < j} Q_{ij} x_i x_j + c$$

Example (for 3 variables):

	x_1	x_2	x_3	Bias (q_i)
x_1	—	Q_{12}	Q_{13}	q_1
x_2	Q_{12}	—	Q_{23}	q_2
x_3	Q_{13}	Q_{23}	—	q_3

Diagonal = linear terms (q_i)

Off-diagonal = quadratic terms (Q_{ij})

3 BINARY TO SPIN MAPPING

Quantum annealers use spin variables $s_i \in \{-1, +1\}$ instead of binary 0/1.

$$s_i = 2x_i - 1 \Leftrightarrow x_i = \frac{1 + s_i}{2}$$

Substitute into $C(x)$ and expand. This transforms the QUBO into an Ising model:

$$C(s) = a + \sum_i h_i s_i + \sum_{i < j} J_{ij} s_i s_j$$

where:

- h_i are local fields (linear terms)
- J_{ij} are couplings (interaction terms) between spins

4 FROM ISING MODEL TO HAMILTONIAN

The Ising model corresponds to the Hamiltonian of the system:

$$H = a'I + \sum_i h_i \sigma_i^z + \sum_{i < j} J_{ij} \sigma_i^z \sigma_j^z$$

where:

- σ_i^z is the Pauli-Z operator for qubit i
- I is the identity operator



The constant term $a'I$ only shifts all energies equally and does not affect the optimal solution.

5 PHYSICAL INTERPRETATION

Each spin configuration $s = (s_1, \dots, s_n)$ has an energy:

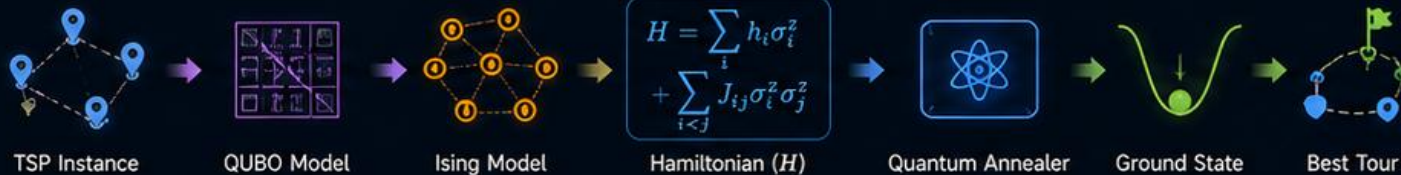
$$E(s) = \langle s | H | s \rangle$$

The quantum annealer searches for the state with **minimum** energy (ground state), which corresponds to the minimum-cost tour of the TSP.



Lowest energy state
= Optimal solution

EXAMPLE WORKFLOW (SUMMARY)



We convert the optimization problem into a physical system whose **lowest energy state** is the optimal answer.



KEY TAKEAWAY

- ✓ QUBO (in binary form) → Ising model (in spin form) → Hamiltonian (quantum system).
- ✓ This mapping allows quantum annealers to use **quantum mechanics** to find the minimum of the objective function.
- ✓ The **ground state** of the Hamiltonian encodes the **optimal TSP tour**.

QUANTUM ANNEALING WORKFLOW FOR TSP

From a Complex Routing Problem to the Best Possible Tour

Quantum annealing finds the lowest-energy state of a problem Hamiltonian. Each step transforms the TSP into a form the quantum annealer can solve.

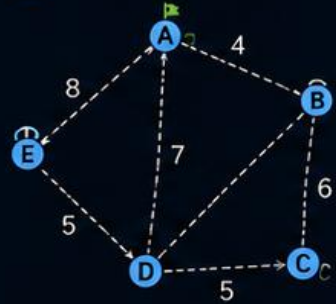


THE GOAL

Find the minimum-cost Hamiltonian cycle that visits each city exactly once and returns to the starting city.



1 TSP INSTANCE



The problem: find the shortest tour that visits each city once and returns to A.

2 MODEL AS QUBO



Encode the problem using binary variables, build the QUBO matrix Q , and define the objective and constraints.

3 MAP TO HAMILTONIAN

$$H = aI + \sum_i h_i \sigma_i^z + \sum_{i < j} J_{ij} \sigma_i^z \sigma_j^z$$



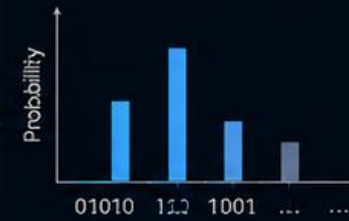
Convert the QUBO into an Ising model (Hamiltonian) with qubits and couplings.

4 QUANTUM ANNEALING



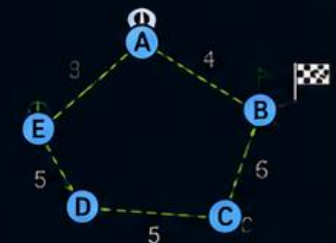
Submit the Hamiltonian to the quantum annealer. It evolves the system from an easy initial state to the ground state (minimum energy).

5 MEASURE OUTPUT



The annealer returns many samples (bitstrings) with associated probabilities. Lower energy = higher quality.

6 DECODE BEST TOUR



Decode the best (lowest-energy) sample into a valid tour. This is the minimum-cost solution (or near-optimal).

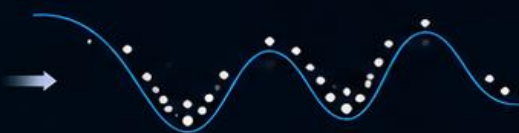
HOW QUANTUM ANNEALING WORKS (IN SIMPLE TERMS)

INITIAL STATE



System starts in a superposition of states (high energy).

ANNEALING PROCESS



The system slowly evolves, exploring many possibilities. It tends to move toward states with lower energy.

FINAL STATE (GROUND STATE)



System ends in the lowest energy state with high probability (optimal solution).

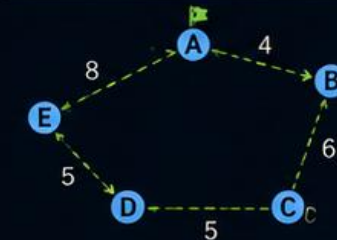


KEY BENEFITS

- ✓ Finds good solutions for hard optimization problems.
- ✓ Naturally handles complex constraints.
- ✓ Explores an enormous solution space in parallel.
- ✓ Ideal for problems like TSP, scheduling, logistics, and more.

EXAMPLE RESULT

Best Tour Found: A → B → C → D → E → A



Total Distance
28



Quantum annealing leverages **quantum mechanics** to find the minimum-energy state of a problem Hamiltonian, providing a powerful approach to solving complex optimization problems like the **Travelling Salesman Problem**.

LIVE EXAMPLE: SOLVING A 6-CITY TSP

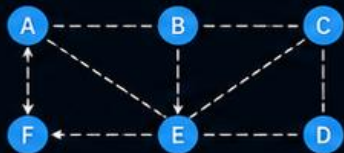
From Cities to Quantum Solution

Let's solve a simple Travelling Salesman Problem with 6 cities using quantum annealing on a D-Wave system.

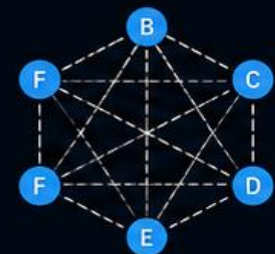


Objective

Find the shortest possible route that visits each city exactly once and returns to the starting city.



1 CITIES



A, B, C, D, E, F
6 cities with pairwise distances

2 DISTANCE MATRIX

	A	B	C	D	E	F
A	0	2	9	10	7	3
B	2	0	6	4	3	8
C	9	6	0	8	5	7
D	10	4	8	0	6	2
E	7	3	5	6	0	4
F	3	8	7	2	4	0

Symmetric distance matrix
(costs between cities)

3 QUBO FORMULATION

We encode the problem using binary variables $x_{i,p}$:
 $x_{i,p} = 1$ if city i is at position p in the tour,
0 otherwise.

Objective:
Minimize the total travel distance
+ penalty terms to enforce constraints.

Minimize $x^T Q x$
subject to:
• Each city appears exactly once
• Each position is occupied by exactly one city

For 6 cities → 36 binary variables
(6 cities × 6 positions)

4 HAMILTONIAN

The QUBO is mapped to an Ising Hamiltonian:

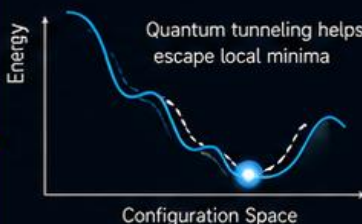
$$H = \sum_i h_i \sigma_i^z + \sum_{i < j} J_{ij} \sigma_i^z \sigma_j^z$$

Where:

- h_i are local fields
- J_{ij} are couplings
- $\sigma_i^z \in \{-1, +1\}$ (Ising spins)

Lower energy states of H
correspond to shorter tours
(feasible solutions).

5 QUANTUM ANNEALING (D-WAVE)



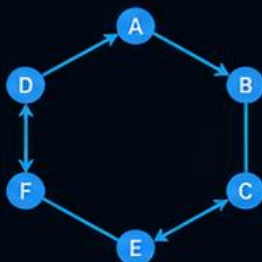
- The system is initialized in a superposition of states.
- Quantum fluctuations explore many solutions in parallel.
- The system is slowly annealed to the problem Hamiltonian.
- The lowest energy state is returned.

6 BEST ROUTE FOUND

Optimal Tour
A → F → D → B → E → C → A

Total Distance
23

Tour Visualization



HOW THE ENCODING LOOKS (Example)

Positions →	1	2	3	4	5	6
City A	1	0	0	0	0	0
City B	0	0	0	1	0	0
City C	0	0	0	0	0	1
City D	0	0	1	0	0	0
City E	0	0	0	0	0	0
City F	0	1	0	0	0	0

Each row (city)
has exactly one 1

Each column (position)
has exactly one 1

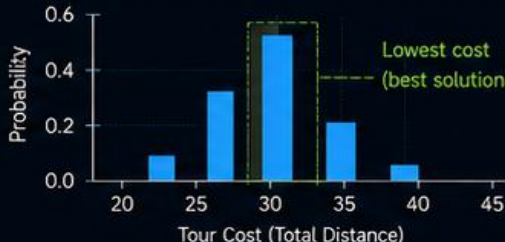


KEY TAKEAWAY FROM THE EXAMPLE

Quantum annealing finds the lowest energy state of the Ising Hamiltonian, which corresponds to the shortest valid tour of the TSP.

For larger problems (50, 100+ cities), this approach can explore a vast number of possibilities efficiently.

RESULTS (D-WAVE SAMPLING)



The best (lowest cost)
solution is returned
with the highest
probability.



This 6-city example demonstrates the complete pipeline:

Cities → Distances → QUBO → Hamiltonian → Annealing → Best Route

The same process scales to real-world problems
with hundreds or thousands of cities.



CASE STUDY: DEUTSCHE BAHN

How Quantum Optimization Can Improve Railway Scheduling

Deutsche Bahn (DB) operates one of Europe's largest rail networks.

Optimizing train routes, maintenance, and schedules is a **massive combinatorial challenge**.



Railway scheduling is like solving thousands of **Travelling Salesman Problems** — every day, in real time, across the entire network.



OVERVIEW

- ✓ DB operates ~33,000 km of tracks in Germany.
- ✓ ~40,000 trains run daily.
- ✓ Millions of connections and transfers across hundreds of stations.
- ✓ Constraints: time windows, track availability, maintenance, delays, passenger demand, safety rules.

HOW TSP APPLIES TO DEUTSCHE BAHN

Consider a set of trains that must visit multiple stations (e.g., for pickups, transfers, or maintenance) and return to their depot — with minimal total time and cost.



 Stations = Cities

 Travel time / distance = Edge cost

 Goal: Find the best route (tour) with minimum total time

QUANTUM OPTIMIZATION POTENTIAL

- ✓ Quantum annealing can evaluate many possible schedules in parallel.
- ✓ Helps find near-optimal solutions faster for huge networks.
- ✓ Ideal for real-time re-optimization when delays or disruptions occur.
- ✓ Enables more efficient use of tracks, trains, and energy.

CLASSICAL vs QUANTUM (Future Potential)

CLASSICAL APPROACH



- ✗ Solves sequentially
- ✗ Struggles with large search spaces
- ✗ Slower for real-time updates

VS

QUANTUM APPROACH



- ✓ Explores many possibilities in parallel
- ✓ Handles larger search spaces efficiently
- ✓ Faster adaptation to real-time changes

THE CHALLENGE

- ✗ Enormous number of possible routes and schedules.
- ✗ Conflicting constraints (tracks, signals, maintenance, platforms).
- ✗ Delays in one part can cascade across the network.
- ✗ Current classical algorithms struggle with real-time, large-scale optimization.

REAL-WORLD IMPACT FOR DEUTSCHE BAHN



Reduced Delays
Better on-time performance



Efficient Resource Usage
Trains, tracks, and energy



Lower Costs & Emissions
More sustainable operations



Better Passenger Experience
Smoother connections and reliability



KEY TAKEAWAY:

Railway scheduling is a **giant optimization problem** with similarities to **TSP**.

Quantum computing, especially quantum annealing, has the potential to **transform** how Deutsche Bahn and other transport systems plan, adapt, and operate in **real time**.



OTHER COMPANIES USING QUANTUM OPTIMIZATION

Real-World Leaders Exploring Quantum Annealing for Complex Optimization Problems

Many global companies are leveraging quantum annealers, especially D-Wave systems, to solve real-world optimization challenges that are hard for classical computers.



Why so many industries?

Quantum annealing excels at combinatorial optimization problems with thousands to millions of variables, which appear in nearly every industry.



VOLKSWAGEN Automotive



Use Case:
Traffic flow optimization for autonomous driving and smart mobility.

Benefit:
Improved routing and reduced congestion in real time.



BMW GROUP Automotive



Use Case:
Factory layout optimization and production scheduling.

Benefit:
Increased efficiency and lower production costs.



AIRBUS Aerospace



Use Case:
Fleet assignment and air traffic management optimization.

Benefit:
Reduced delays and optimized fuel consumption.

DENSO Mobility & Manufacturing



Use Case:
Optimization of logistics, scheduling, and component placement.

Benefit:
More efficient operations and reduced energy usage.



LOCKHEED MARTIN Defense & Aerospace



Use Case:
Resource allocation and complex mission planning.

Benefit:
Better decision-making under uncertainty and constraints.



TOYOTA Automotive



Use Case:
Supply chain optimization and parts distribution.

Benefit:
Lower operational costs and improved delivery times.



ROCHE Healthcare & Pharma



Use Case:
Molecular docking and drug discovery optimization.

Benefit:
Faster discovery of potential drug candidates.

JPMORGAN CHASE & Co. Finance



Use Case:
Portfolio optimization and risk analysis.

Benefit:
Better risk-adjusted returns and faster scenario analysis.



Quantum optimization is driving real impact across industries
From mobility and manufacturing to healthcare and finance.



Common Goal:
Solve bigger problems, faster and more efficiently, with lower cost and higher accuracy.



REAL INDUSTRIAL WORKFLOW

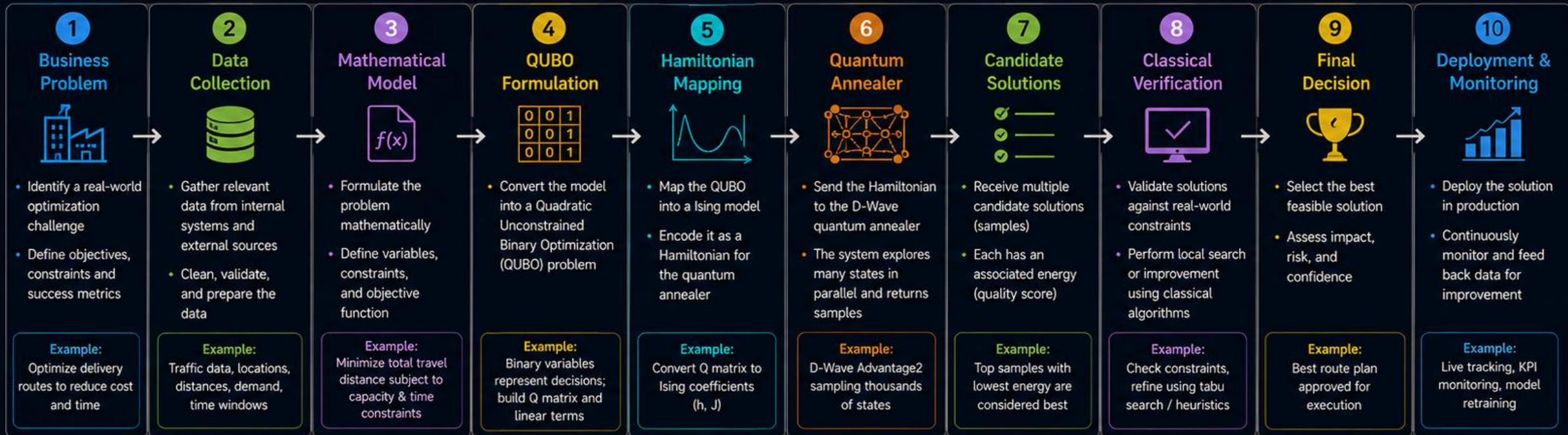
From Business Problem to Optimized Decision

Quantum optimization is not just about hardware. It is a complete workflow that combines domain knowledge, mathematical modeling, quantum computing, and classical post-processing to deliver real-world impact.



The Goal

Solve complex optimization problems that are hard for classical computers, and deliver actionable solutions for real-world operations.



WHY THIS WORKFLOW MATTERS



Combines domain expertise with quantum power



Delivers better solutions for hard optimization problems



Leverages the best of quantum and classical computing



Ensures solutions are practical, feasible, and reliable



Creates a feedback loop for continuous improvement

REAL-WORLD IMPACT



Logistics: Lower cost, faster delivery



Manufacturing: Better scheduling



Finance: Portfolio optimization



Energy: Grid optimization



Aviation: Crew & fleet planning



Healthcare: Resource allocation

CURRENT LIMITATIONS

Where Quantum Annealing Stands Today

Quantum annealing shows great promise for optimization problems like TSP, but several technical and practical limitations remain.



THE REALITY

Quantum annealing is a powerful tool, but it is not a silver bullet—yet. Ongoing research continues to push the boundaries.



1 LIMITED PROBLEM SIZE



- Current annealers have a limited number of qubits (e.g., ~5,000 usable qubits on Advantage).
- Mapping large TSP instances requires many qubits.
- Practical problem sizes are still relatively small (typically up to 100–200 cities with embedding).

2 EMBEDDING OVERHEAD



- The hardware has sparse qubit connectivity (e.g., Pegasus topology).
- Logical qubits must be chained together, increasing the number of physical qubits used.
- This increases complexity and reduces solution quality.

3 NO GUARANTEE OF OPTIMALITY



- Quantum annealing is a heuristic approach.
- It returns good (low-energy) solutions but not always the optimal solution.
- Multiple runs may be needed, with no certainty of finding the global optimum.

4 SOLUTION QUALITY & CONSISTENCY



- Results can vary between runs due to quantum noise.
- Probability of finding the best solution may be low for hard instances.
- Requires careful parameter tuning (annealing time, chains, penalties).

5 NOISE AND DECOHERENCE



- Quantum systems are highly sensitive to noise and thermal fluctuations.
- Errors can lead to incorrect solutions.
- Error correction is still limited on current hardware.

6 ACCESSIBILITY AND COST



- Access to quantum annealers is limited (cloud-based, queue times).
- Commercial systems are expensive to build and maintain.
- Not always practical for real-time or large-scale industrial use—yet.



BOTTOM LINE

Quantum annealing is not a replacement for classical optimization—yet. It is a complementary approach that excels on specific problem types and continues to improve rapidly.

WHAT'S NEXT?



More qubits with better connectivity



Better error correction and noise reduction



Hybrid quantum-classical algorithms



Wider adoption and real-world integration



The future is bright: As hardware and algorithms evolve, quantum annealing will tackle **larger, more complex problems**—bringing us closer to **real-world impact**.

RESEARCH CHALLENGES

Key Hurdles on the Path to Practical Quantum Advantage

Quantum computing is advancing rapidly, but several fundamental challenges remain before it can solve large-scale, real-world problems reliably and efficiently.



The Road Ahead

Overcoming these challenges requires breakthroughs in physics, engineering, algorithms, and software. Progress in each area brings us closer to practical, scalable, and fault-tolerant quantum systems.

1 BUILDING BETTER QUBITS

Quality over quantity



- Qubits are extremely sensitive to noise and environment.
- Improving coherence time (longer stability) is critical.
- Developing qubits with lower error rates and higher fidelity remains a major challenge.

2 ERROR CORRECTION & MITIGATION

Fighting errors at scale



- Qubits are prone to errors from decoherence and operational imperfections.
- Quantum error correction requires many physical qubits per logical qubit.
- Efficient, scalable error correction is still a work in progress.

3 SCALABILITY

From dozens to thousands (and beyond)



- Current systems have limited number of qubits and connectivity.
- Scaling up while maintaining performance is a major engineering challenge.
- We need systems with millions of qubits to solve truly impactful problems.

4 EMBEDDING COMPLEX PROBLEMS

Mapping to hardware is not easy



- Real-world problems often require complex connections between variables.
- Hardware has limited connectivity, so logical qubits must be “embedded” using chains of physical qubits.
- Better embedding techniques can significantly improve solution quality and scale.

5 QUANTUM ADVANTAGE BENCHMARKS

Proving real-world benefit



- It's still challenging to demonstrate clear quantum advantage over classical supercomputers.
- We need better benchmarks and real-world use cases where quantum clearly wins in time, cost, or accuracy.

6 SOFTWARE & ALGORITHM DEVELOPMENT

The brain behind the quantum power



- New algorithms and formulations are needed to fully exploit quantum hardware.
- Hybrid quantum-classical approaches and better tools can unlock near-term practical value.



Summary

Research challenges are not roadblocks—they are opportunities.

Solving them will unlock the full potential of quantum computing and drive the next technological revolution.



FUTURE OUTLOOK

Quantum Annealing is Evolving — and So Are the Possibilities

While still in its early days, quantum annealing is rapidly advancing.

The next decade will unlock greater power, accessibility, and real-world impact.



THE VISION

A future where quantum optimization works hand-in-hand with classical systems to solve the world's most complex problems—faster, greener, and smarter.



1. HARDWARE ADVANCES

- More qubits with higher connectivity
- Improved qubit stability and lower noise
- New generations of annealers (e.g., Advantage2 and beyond)



More power and precision will enable larger and more complex problems.



2. HYBRID & INTEGRATED SOLUTIONS

- Deeper integration with classical algorithms and HPC
- Smarter hybrid workflows
- Tailored to specific industries and use cases



Best of both worlds: quantum speed + classical versatility.

WHAT'S AHEAD?



3. CLOUD ACCESS & SCALABILITY

- More cloud platforms and services making quantum accessible
- Pay-as-you-go models
- Lower barrier to entry for businesses and researchers



Wider adoption across industries and organizations of all sizes.



4. NEW ALGORITHMS & APPLICATIONS

- Problem-specific algorithms
- New formulations beyond QUBO
- Expansion into areas like finance, energy, healthcare, and climate modeling



Unlocking value in more industries and solving new classes of problems.



5. TOWARD QUANTUM ADVANTAGE

- Reaching problems where quantum annealing clearly outperforms classical methods
- Demonstrating real-world quantum advantage



From promise to proof: delivering measurable, transformational impact.

POTENTIAL IMPACT AREAS



Transportation

Smarter routing and scheduling



Energy

Grid optimization and efficiency



Logistics

Supply chain and fleet optimization



Finance

Portfolio optimization and risk analysis



Healthcare

Resource allocation and drug discovery



Environment

Climate modeling and sustainability

THE ROAD AHEAD



Today

Early adoption and year-over-year improvements



Next 3–5 Years

Broader access, better performance, more use cases



Next 5–10 Years

Mature hybrid systems, industry integration, growing impact



Beyond 10 Years

Quantum advantage in real-world optimization



The future of optimization is quantum-enhanced. **Together**, we can **solve more, faster**, and build a **better world**.

KEY TAKEAWAYS

Quantum Annealing for Combinatorial Optimization

Quantum annealing offers a novel and powerful approach to solving complex optimization problems like the Travelling Salesman Problem (TSP).



THE BIG PICTURE

We are at the beginning of a quantum-powered era of optimization — solving bigger problems, smarter, and more efficiently.



1 A POWERFUL ALTERNATIVE



Quantum annealing leverages quantum mechanics to find low-energy (near-optimal) solutions for hard combinatorial problems like TSP.

- ✓ It explores many possibilities in parallel.

2 FROM PROBLEM TO SOLUTION



We transform TSP into a QUBO model, map it to a Hamiltonian, and use a quantum annealer to search for the minimum energy state.

- ✓ A structured pipeline turns a real-world problem into a quantum-ready model.

3 REAL-WORLD IMPACT



Industries like Deutsche Bahn can benefit from optimized routing and scheduling — leading to reduced delays, lower costs, and improved efficiency.

- ✓ Quantum optimization has practical value today.

4 PROMISING POTENTIAL



Quantum annealing is already delivering useful results and continues to improve with better hardware, algorithms, and hybrid approaches.

- ✓ We are moving from promise to performance.

5 CURRENT LIMITATIONS



Challenges remain: limited qubit connectivity, noise, solution quality, and cost still restrict large-scale applications.

- ✓ Awareness of limitations helps guide responsible innovation.

6 THE ROAD AHEAD



The future holds more qubits, better algorithms, broader access, and real-world quantum advantage across industries.

- ✓ Quantum annealing will play a key role in solving tomorrow's problems.



IN SUMMARY

Quantum annealing is a complementary technology that expands our optimization toolkit. It will not replace classical methods—but together, they will solve more, faster.



THE BOTTOM LINE

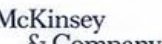
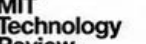
Quantum annealing brings us **one step closer** to solving the world's most complex optimization challenges — **efficiently and at scale**.

“ Quantum annealing isn't just about qubits — it's about solving meaningful problems and creating **real impact**. ”

REFERENCES



Key sources and further reading for quantum computing, quantum annealing, and industrial applications

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2		D-Wave Systems Inc. <i>Quantum Annealing: An Introduction and Practical Guide</i> https://docs.dwavesys.com
3		IBM Quantum <i>What is Quantum Computing?</i> https://www.ibm.com/quantum/learn/what-is-quantum-computing
4		IBM Quantum <i>Quantum Algorithms and Applications</i> https://www.ibm.com/quantum/learn/algorithms
5		Google Quantum AI <i>Quantum Computing Explained</i> https://quantumai.google/learn
6		Cerezo, M. et al. <i>Variational quantum algorithms.</i> <i>Nature Reviews Physics</i> , 3, 625–644 (2021). https://doi.org/10.1038/s42254-021-00348-9
7		Arute, F. et al. <i>Quantum supremacy using a programmable superconducting processor.</i> <i>Science</i> , 5749, 1084–1089 (2019). https://doi.org/10.1126/science.aag4963
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9		Huang, H. et al. <i>A survey of optimization with quantum annealing.</i> <i>Springer, Annals of Operations Research</i> , 314, 141–175 (2022). https://doi.org/10.1007/s10479-021-04178-1
10		McKinsey & Company <i>Quantum Technology Monitor: Trends and Outlook.</i> https://www.mckinsey.com/capabilities/mckinsey-quantumblack
11		Bain & Company <i>Quantum Technology Report 2023.</i> https://www.bain.com/insights/quantum-technology-report
12		Boston Consulting Group <i>The Quantum Advantage: How Quantum Computing Can Solve Industrial Problems.</i> https://www.bcg.com/publications/quantum-advantage
13		Accenture <i>Quantum Computing: The Next Mega Technology.</i> https://www.accenture.com/quantum-computing
14		QUTIS <i>Applications of Quantum Annealing.</i> https://qut.is/applications
15		Los Alamos National Laboratory <i>Introduction to Quantum Annealing.</i> https://www.lanl.gov/science-innovation/quantum
16		Rigetti Computing <i>An Introduction to Quantum Algorithms.</i> https://www.rigetti.com/learn
17		MIT Technology Review <i>The Race to Commercial Quantum Computing.</i> https://www.technologyreview.com/
18		Deutsche Bahn AG <i>Using Quantum Annealing for Railway Optimization.</i> https://www.deutschebahn.com/en/innovation
19		Volkswagen Group <i>Quantum Computing in Logistics and Production Planning.</i> https://www.volkswagen-newsroom.com
20		DENSO <i>Quantum-Inspired Optimization for Mobility.</i> https://www.denso.com/global/en/innovation

ADDITIONAL RESOURCES



D-Wave Leap™
Access to D-Wave quantum computers in the cloud.
<https://cloud.dwavesys.com>



Qiskit Textbook
Open-source quantum computing textbook by IBM.
<https://qiskit.org/textbook>



Microsoft Quantum Development Kit
Open-source tools for quantum algorithm development.
<https://quantum.microsoft.com>



Quantum Computing Report (BCG)
Detailed insights and case studies.
<https://www.bcg.com>



Quantum Country Profiles (McKinsey)
Global overview of quantum initiatives and investments.
<https://www.mckinsey.com>

“This field is evolving rapidly.
Stay curious and keep exploring!”



QUESTIONS & DISCUSSION

Let's Explore Ideas, Challenges, and the Future Together

Quantum computing is a rapidly evolving field with tremendous potential. Your questions, insights, and perspectives are valuable!

“

The best way to predict the future is to invent it.

– Alan Kay



DISCUSSION PROMPTS



1. THE BIG PICTURE

- Will quantum computers replace classical computers, or will both coexist?
- Which industries do you think will benefit first and the most from quantum optimization?



2. QUANTUM ADVANTAGE

- Is quantum advantage already here, or are we still in the early days?
- What types of problems are best suited for quantum annealing today?



3. TECHNOLOGY PERSPECTIVE

- What technical breakthroughs are most needed to scale quantum systems?
- How important are error correction and qubit quality for real-world impact?



4. BUSINESS & IMPACT

- What business problems in your field could be solved using quantum optimization?
- How can organizations start preparing for the quantum era today?



5. CHALLENGES & LIMITATIONS

- What do you see as the biggest barriers to adoption right now?
- How do we balance hype with realistic expectations?



6. THE FUTURE

- Where do you see quantum computing in 10–20 years?
- What would be the “killer application” that changes the world?



OPEN FLOOR

What are your thoughts, questions, or ideas?
Let's learn and explore together!



LET'S CONNECT THE DOTS

Quantum computing is not just about technology—it's about solving the world's most important problems in new and powerful ways.

Where will we go from here?



THANK YOU FOR YOUR ATTENTION! I LOOK FORWARD TO YOUR **QUESTIONS** AND A GREAT **DISCUSSION**.





ECONOMIC IMPACT OF QUANTUM COMPUTING

Quantum computing is not just a technology breakthrough—
it is an **economic catalyst** for the next decade and beyond.

“ Quantum computing has the potential
to unlock **\$850B – \$1.3T** in global
economic value by 2035.” ”
– McKinsey Global Institute



KEY ECONOMIC IMPACTS



Boosts Productivity & Efficiency

Solves complex optimization problems faster,
reducing time, cost, and resource usage.



Cost Savings

Optimizes operations, logistics, energy, and
finance to significantly cut operational costs.



Drives Innovation

Enables breakthroughs in materials, drug discovery,
AI, and climate modeling.



Creates New Markets & Jobs

Fueling a new quantum economy with high-value
industries, startups, and skilled jobs.



Strengthens Global Competitiveness

Countries and companies investing early will lead
the next wave of technological leadership.



MARKET & GROWTH OUTLOOK

Global Quantum Computing Market Size

Market Size (USD)



The global quantum computing market is expected to reach
\$25.1B by 2035, growing at a **CAGR** of **46%**.



ECONOMIC BENEFITS ACROSS SECTORS



SUPPLY CHAIN & LOGISTICS

Optimized routing and scheduling
can reduce logistics costs by
10–20%.



FINANCIAL SERVICES

Better risk analysis and portfolio
optimization could add
\$200–400B in value globally.



ENERGY & UTILITIES

Smarter grid management and
energy optimization could save
\$60–90B annually.



PHARMA & HEALTHCARE

Faster drug discovery and
personalized medicine could
unlock **\$100–200B** in value.

INDUSTRIES THAT WILL BENEFIT THE MOST



Manufacturing



Logistics



Finance



Energy



Healthcare



Chemistry



Cybersecurity



Environment



Public Sector



QUANTUM COMPUTING IS MORE THAN A TECHNOLOGICAL ADVANCEMENT – IT IS AN **ECONOMIC TRANSFORMATION**.



INVEST TODAY. LEAD TOMORROW.



ECONOMIC BENEFITS OF QUANTUM OPTIMIZATION

Quantum optimization enables organizations to solve complex problems more efficiently, creating **measurable economic value** across industries.



1. COST SAVINGS

Optimize resources, reduce waste, and lower operational costs.



Impact: **5–30%** reduction in operational costs



2. REVENUE GROWTH

Better decisions lead to improved service quality, more customers, and higher revenue.



Impact: **3–10%** increase in revenue potential



3. OPERATIONAL EFFICIENCY

Solve complex scheduling, routing, and planning problems faster and more accurately.



Impact: **10–40%** improvement in efficiency



4. INNOVATION & COMPETITIVE ADVANTAGE

Early adoption positions companies as technology leaders and innovation pioneers.



Impact: **Stronger market position and brand differentiation**



5. SUSTAINABILITY

Optimize energy usage, reduce emissions, and support greener supply chains.



Impact: **Lower carbon footprint and energy costs**



6. RISK REDUCTION

Improve risk analysis and create more resilient, adaptable business strategies.



Impact: **Reduced risk exposure and better compliance**

REAL-WORLD ECONOMIC POTENTIAL



COST REDUCTIONS

\$60–90B

annual potential in energy & utilities alone



SUPPLY CHAIN OPTIMIZATION

10–20%

logistics cost reduction



HEALTHCARE SAVINGS

\$100–200B

in drug discovery & personalized medicine



GLOBAL MARKET OPPORTUNITY

\$850B – \$1.3T

in economic value by 2035*

*McKinsey Global Institute

“ Quantum optimization is not just about solving hard problems—it’s about creating **real economic value** and a sustainable future. ”

– Boston Consulting Group (BCG)

INDUSTRIES THAT BENEFIT THE MOST



Logistics & Transportation



Manufacturing



Energy & Utilities



Financial Services



Healthcare & Pharmaceuticals



Retail & E-commerce



Telecom & Networks



Aerospace & Defense

GLOBAL INVESTMENT IN QUANTUM TECHNOLOGY

A worldwide race to lead the **next computing revolution**



Quantum technology is a **strategic priority** nations and industries are investing billions to secure the quantum future.

MAJOR NATIONAL INITIATIVES & INVESTMENTS



UNITED STATES

National Quantum Initiative Act (2018)
\$1.2B+
federal funding committed (2018–2023)



EUROPEAN UNION

Quantum Flagship Program (2018)
€1B+
committed until 2028 across member states



GERMANY

Quantum Technologies Program
€2B+
by 2026 to strengthen research, industry & startups



CHINA

National Laboratory for Quantum Information Sciences
\$15B+
estimated investment over the next decade



JAPAN

Quantum Technology Innovation Strategy (2019)
¥800B+
(~\$6B+) through 2030



CANADA

National Quantum Strategy
\$360M+
over 5 years (2019–2024)

WHY THE WORLD IS INVESTING



STRATEGIC LEADERSHIP

Quantum technology will drive future economic and military leadership.



ECONOMIC OPPORTUNITY

Potential to unlock trillions in value across industries and create new markets.



TECHNOLOGICAL SOVEREIGNTY

Reduce dependence on foreign technology and secure national innovation.



TALENT & INNOVATION ECOSYSTEM

Attract top talent, build startups, and drive research breakthroughs.



LONG-TERM IMPACT

Quantum will revolutionize computing, communications, security, and sensing.



KEY TAKEAWAY

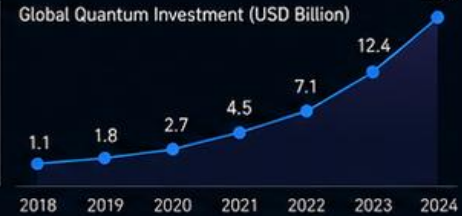
Quantum technology is the next frontier of innovation. Countries that invest today will **lead the industries of tomorrow.**

GLOBAL INVESTMENT TRENDS



Global investment in quantum technologies exceeded
\$35B+
in private & public funding (2018–2024)

Source: McKinsey, 2023



WHO IS INVESTING?



Governments & Public Agencies



Tech Giants
(Google, IBM, Microsoft, Amazon)



Startups
(IonQ, Rigetti, PsiQuantum, etc.)



Universities & Research Labs



Venture Capital & Private Investors



The quantum race is not just about technology—it's about **economic growth, national security, and shaping the future.**